

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
4.	(a)			Increases in mass values are equal to the increases in volume values / every 10 [cm ³] increases the mass by 10 [g] [so correct] Accept correct calculation so $\frac{10}{10} = 1$ Accept when one increases by 10 the other goes up by 10 or they both go up in equal steps			1	1	1	1
	(b)	(i)		Accept in the range 40 - 50 [cm ³]		1		1		1
		(ii)		The mass values stop increasing by 10 [after 40 cm ³] or increases in mass changes [beyond 40 cm ³]		1		1	1	1
	(c)			Density = $\frac{79}{10}$ or $\frac{158}{20}$ or $\frac{237}{30}$ etc or $\frac{(424 - 108)}{(80 - 40)}$ (1) 7.9 [g/cm ³] (1) Award 1 mark for correct substitution of mass and linked volume from the table N.B. volume must be above 40		2		2	2	2
	(d)			Better resolution on the measuring cylinder (1) Balance with better resolution / to more dp's (1) Accept smaller ball bearings or more precise balance / measuring cylinder Don't accept repeat the experiment or more accurate equipment			2	2		2
				Question 4 total	0	4	3	7	4	7